Range

1

a 13 b 31 c 2.2981

2

a 5 b 72

3 4

4 23

5 11

6

a One person in the group

b 15 people

c 16 names

d 10 names

e 6

Interquartile range

7

a

i 7 ii 13 iii 15 iv 7

v 16 vi 9

b

i 7 ii 1 iii 18 iv -3

v 9 vi 12

c

i 7 ii 38 iii 38 iv 33

v 48 vi 15

d

i 7 ii 2.9 iii 3.6 iv 1.6

v 3.5 vi 1.9

e

i 15 ii 35 iii 27 iv 27

v 40 vi 13

f

i 19 ii 28 iii 43 iv 13

v 38 vi 25

8

a

i 15   ii 9   iii 16   iv 7

b

i 17.5   ii 45   iii 27.5   iv 16

c

i 25   ii 17.5   iii 45   iv 27.5

d

i 25   ii 19.5   iii 26.5   iv 7

9

a

i 9   ii 10   iii 1

b

i 1   ii 3   iii 2

c

i 72   ii 85   iii 13

10

a \$15   **b** \$16.50   **c** International

11

a 47   **b** 7   **c** 4

12

a 51   **b** 7   **c** 1

13

a

Class	Class center	Frequency
1 – 5	3	5
6 – 10	8	7
11 – 15	13	6
16 – 20	18	10

b

i 15   ii 10

14

a 75   **b** 8   **c** 3

15 8.8

16 50

17

a

Ball speed (km/h)	Class center	Frequency
$110 \leq x < 120$	115	4
$120 \leq x < 130$	125	12
$130 \leq x < 140$	135	8
$140 \leq x < 150$	145	5
$150 \leq x < 160$	155	2

b

i 40 ii 125 iii 135 iv 10

18

a

Height (cm)	Class center	Frequency
$95 \leq x < 100$	97.5	1
$100 \leq x < 105$	102.5	3
$105 \leq x < 110$	107.5	2
$110 \leq x < 115$	112.5	12
$115 \leq x < 120$	117.5	18
$120 \leq x < 125$	122.5	15
$125 \leq x < 130$	127.5	4
$130 \leq x < 135$	132.5	4

b

i 35 ii 10

19

a

i 86 **ii** 109 **iii** 23 **iv** 96

v 107 **vi** 11

b

Weight (g)	Class center	Number of apples
$85 \leq x < 90$	87.5	2
$90 \leq x < 95$	92.5	3
$95 \leq x < 100$	97.5	5
$100 \leq x < 105$	102.5	7
$105 \leq x < 110$	107.5	13

c

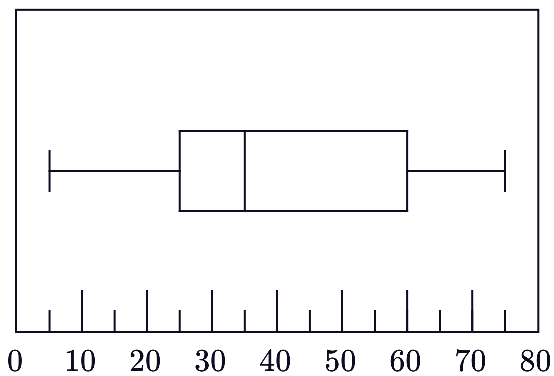
i 20 **ii** 10

d 1

Box plots

20 7

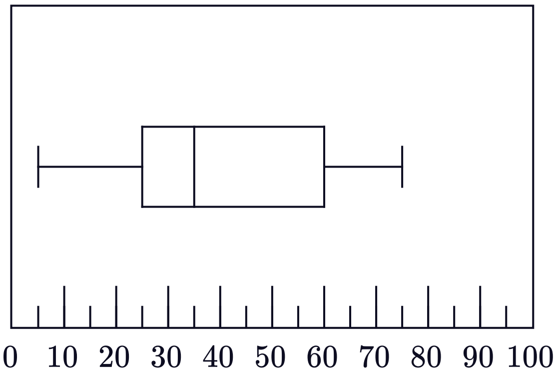
21



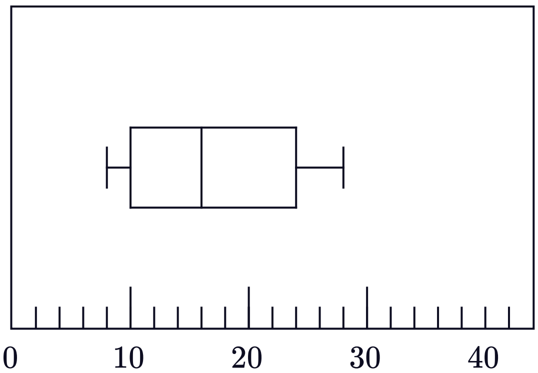
22 33

23 42

24



25

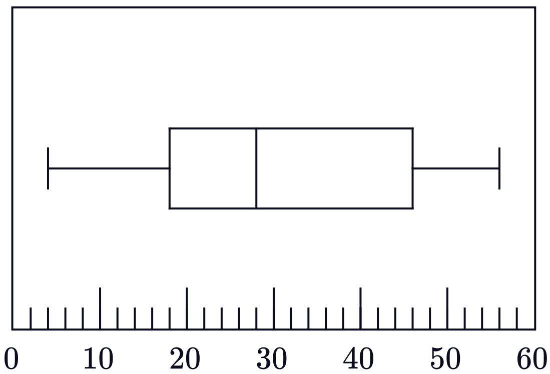


26

a

Minimum	4
Lower Quartile	18
Median	28
Upper Quartile	46
Maximum	56

b



27

a 18.97 kg

b

Q1	17 kg
Median	19 kg
Q3	21 kg

c 2nd

28

a 3

b 18

c 15

d 10

e 7

29

a

i 50% ii 25% iii 50% iv 75%

v 75%

b 2nd

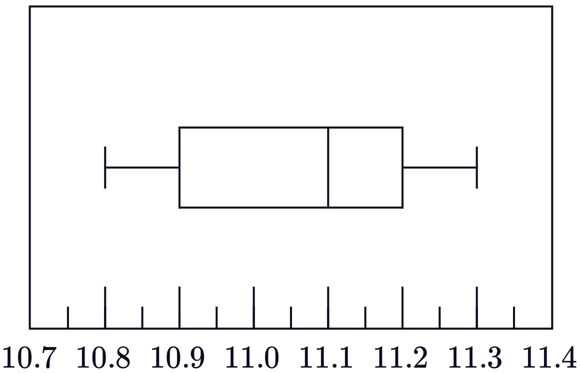
30

a 11.10 mm

b 0.3 mm

c

Glass width



d 78.95%

e 2nd

f The mode and mean

31 No. For less than 25% of the time, he blacks out for longer than 7 seconds.

32

a 31 years b 17 years c 25% d 7

e 1st f 5

Standard deviation

33

a

i 5.30 ii 5.58

b

i 9.57 ii 10.15

c

i 2.01 ii 2.02

d

i 18.74 ii 19.51

e

i 1.58 ii 1.59

f

i 1.58 ii 1.58

g

i 1.61 ii 1.65

h

i 1.36 ii 1.38

34

a 0.56

b
i Increase
ii Increase

c 0

35

a 0.57

b Decrease the mean and increase the standard deviation.

36 Decrease both the mean and standard deviation.

37

a 12

b 9

c 4.7

d Standard deviation. For an accurate representation of spread, every score should be taken into account.

38 The mean for Paper 1 would be greater and the standard deviation would be smaller than that of Paper 2.

39

a $\$5404$ million

b greater

c greater

40

a 16

b 9

c

Score (x)	$x - \text{mean}$	$(x - \text{mean})^2$
19	3	9
18	2	4
14	-2	4
19	3	9
10	-6	36

d 3.94

41

a 11

b 56

c

Score (x)	$x - \text{mean}$	$(x - \text{mean})^2$
-15	-26	676
-7	-18	324
10	-1	1
26	15	225
41	30	900

d 23.05